

Requirement Analysis

Domain:Event Management and Coordination

Problem statement: Event Registration and Attendee Management platform

Course: B.TECH - Computer Science & Engineering

Institution: IIIT SRICITY

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PREFACE

Event registration and attendee management play a crucial role in the success of any event. In many cases, these processes are handled manually or through disconnected tools, which often results in miscommunication, long queues, data errors, and difficulty in tracking participants. As the scale and complexity of events increase, there is a clear need for a unified digital platform that can efficiently manage registrations, attendee information, and on-site verification.

The **Event Registration and Attendee Management Platform** is proposed to address these challenges by providing a centralized, web-based system that supports the complete attendee lifecycle—from registration to event participation. This Software Requirements Specification (SRS) document outlines the requirements and expected behavior of the system in a clear and structured manner. It serves as a reference to understand what the system is intended to do before development begins.

The platform is designed to support different user roles, including clients, event managers, event staff, and attendees. Each role interacts with the system based on specific responsibilities such as event setup, registration management, attendee verification, and reporting. Key functionalities of the system include online event registration, secure payment processing, QR code-based attendee verification, real-time attendance tracking, and notification services.

This document follows standard software engineering practices and focuses on defining both functional and non-functional requirements of the system. The requirements described here will guide the design and development phases in the upcoming semesters and help ensure that the final system is reliable, user-friendly, and aligned with real-world event management needs.

Introduction

The purpose of this requirement analysis document is to clearly identify, analyze, and document the functional and non-functional requirements of the **Event Registration and Attendee Management Platform**. This document serves as a formal agreement between stakeholders and developers and provides a foundation for system design, development, testing, and validation.

Scope of the Project

The Event Registration and Attendee Management Platform provides a web-based solution for managing events, registering attendees, and tracking participation. The system enables attendees to register for events, event managers to manage registrations and attendees, and sponsors to approve events. It automates the event lifecycle while ensuring secure access and efficient handling of event and attendee information.

Definitions

Domain Term	Meaning
Attendee	A user who registers for an event
Event sponsor(client)	Requests and Approves events
Event Manager	Manages events,staff and attendees
Event Staff	Responsible for verifying attendees, managing check-in, supporting on-site activities, and ensuring smooth event execution.
System Admin	Manages, secures, and maintains the platform and its users.
Notification Service	Sends alerts and updates to users about events, registrations, and system activities.

Payment	The transfer of money made by an attendee to complete event registration and confirm participation in an event.
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Description

Product Perspective:

The Event Registration and Attendance Management Platform is a **standalone web-based application** designed to simplify the process of event organization and attendee management. It serves as a centralized platform connecting attendees, event organizers, staff, and system administrators, eliminating the need for manual record-keeping. The system may integrate with external services like **email/SMS gateways** for notifications and **payment gateways** for paid events.

User Classes & Characteristics

User Type	Description
Event sponsor(client)	An individual or organization that requests, funds, or approves an event and provides requirements for its execution.
Attendee	Registers for events, views event details, receives notifications, and manages their profile.
Event Manager	Creates and manages events, views attendee lists, manages staff, generates reports, and communicates with attendees.
Event Staff	Assists organizers in managing attendees, verifying registrations, and event logistics.
System Admin	Manages platform settings, user accounts, security, and ensures smooth system operation.

Operating Environment

The platform is a web-based system accessible through modern browsers and mobile devices. It is hosted on cloud or on-premise servers and requires internet connectivity for real-time event registration, attendee management, and notifications. Emails and SMS are used to communicate updates and alerts. The system supports multiple simultaneous users while maintaining performance and reliability. It provides a responsive interface suitable for both organizers and attendees.

User Requirements

Attendee requirements:

- The attendee should be able to register and log in to the system.
- The attendee should be able to view and search available events.
- The attendee should be able to register for events.
- The attendee should receive notifications and event updates.
- The attendee should be able to manage personal profile details.

Event Manager (Event Organizer) requirements:

- The event manager should be able to create, update, and cancel events.

- The event manager should be able to view and manage attendee registrations.
- The event manager should be able to send notifications to attendees.
- The event manager should be able to generate event-related reports.

Event Staff requirements:

- Event staff should be able to view assigned events.
- Event staff should be able to verify attendee registrations.
- Event staff should assist in attendee check-in and coordination.
- Event staff should update attendee status during events.

Event Sponsor Requirement:

- Event sponsors should be able to register and log in to the platform.
- Event sponsors should be able to request event creation by submitting requirements to the event manager.
- Event sponsors should be able to approve events and provide sponsorship funds.
- Event sponsors should be able to view the status of sponsored events (approved, ongoing, completed).
- Event sponsors should be able to access reports related to sponsored events, including attendee participation.

System Administrator requirements:

- The system administrator should be able to manage user accounts and roles.
- The system administrator should monitor system performance and security.
- The system administrator should manage data backups and recovery.
- The system administrator should configure system settings and integrations.

System Requirements

Functional Requirements

1. User Authentication and Authorization

- The system should allow users to register and log in securely.
- The system should support role-based access for attendees, event managers, staff, and administrators.

2. Event Management

- The system should allow event managers to create, update, and delete events.
- The system should allow event managers to publish and cancel events.

3. Attendee Registration

- The system should allow attendees to view and search available events.
- The system should allow attendees to register and cancel event registrations.
- The system should track attendee registration status.

4. Attendee Management

- The system should allow event managers to view and manage attendee lists.
- The system should allow event staff to verify attendee registrations.
- The system should support attendee check-in during events.

5. Notification Management

- The system should send notifications for registration confirmation, reminders, and event updates.
- The system should support email and SMS notification services.

6. Reporting and Analytics

- The system should generate reports on events and attendee participation.
- The system should allow authorized users to view and download reports.

7. System Administration

- The system should allow administrators to manage users and roles.
- The system should allow administrators to configure system settings.

- The system should monitor system activities and logs.

Non-Functional Requirements

1. Performance

- The system should support multiple concurrent users.
- The system should respond to user requests within acceptable time limits.

2. Security

- The system should ensure secure authentication and authorization.
- The system should protect sensitive data through encryption.

3. Usability

- The system should provide a user-friendly and intuitive interface.
- The system should be accessible on both desktop and mobile devices.

4. Reliability

- The system should operate with minimal downtime.
- The system should recover gracefully from system failures.

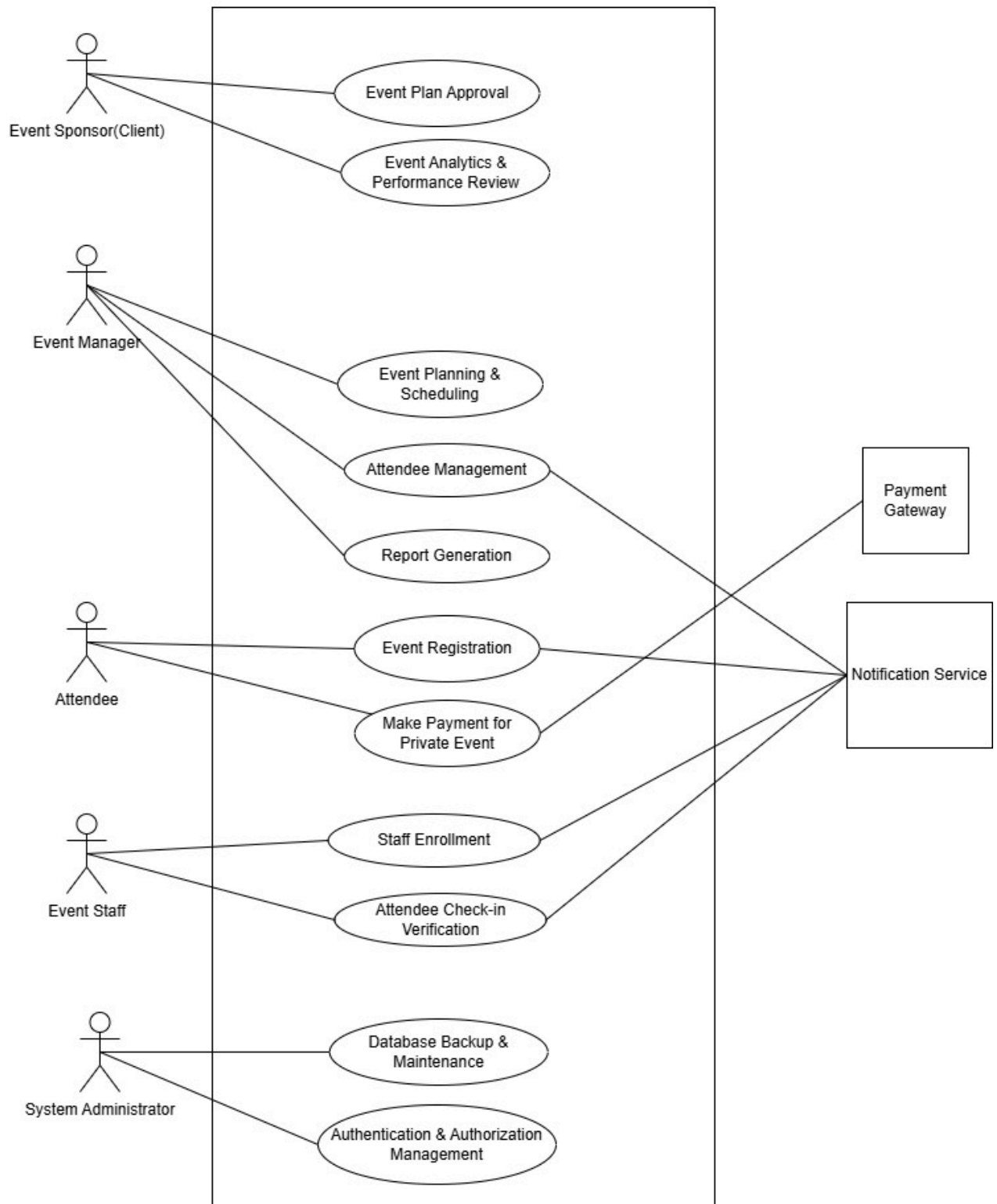
5. Scalability

- The system should handle growth in the number of users and events.

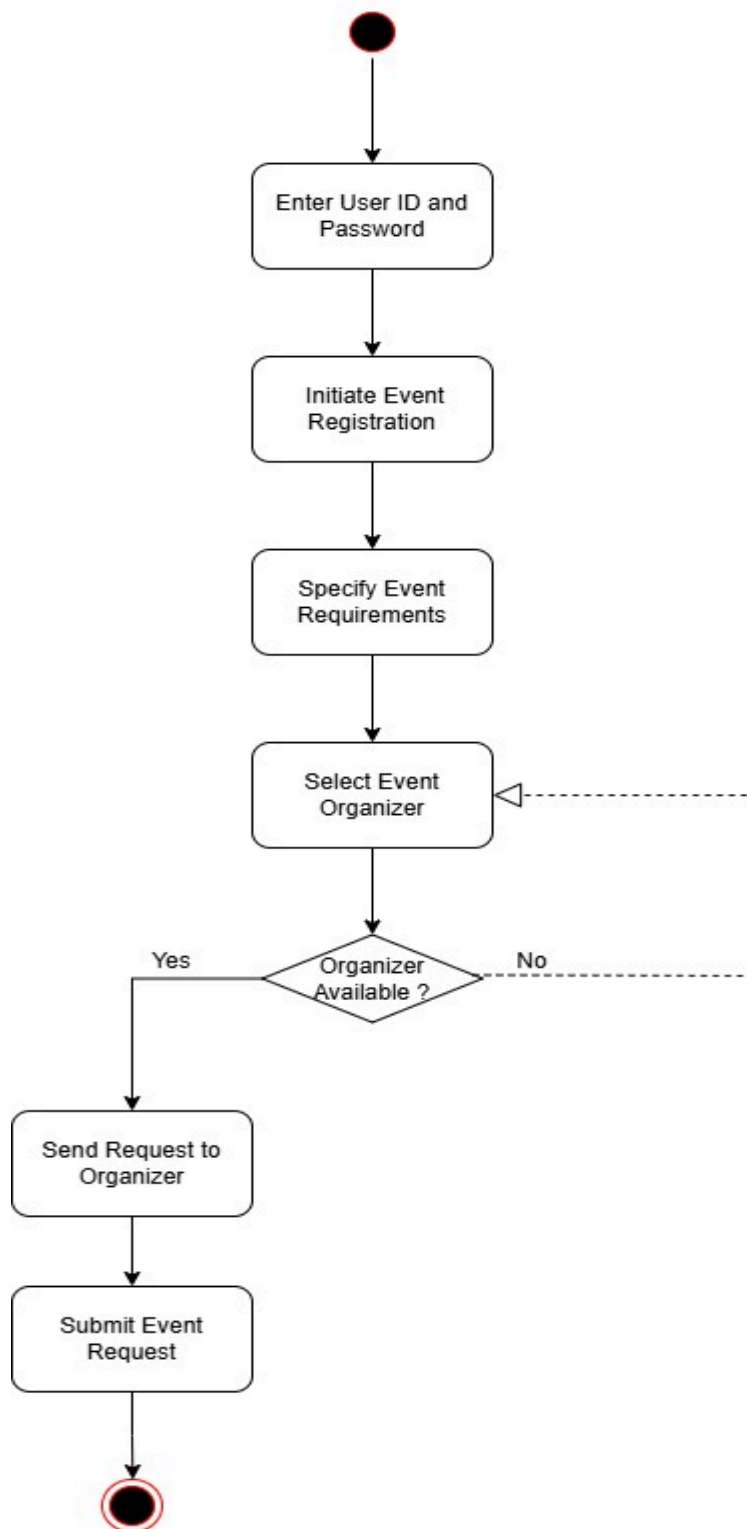
6. Maintainability

- The system should allow easy updates and maintenance.
- The system should support logging and error monitoring.

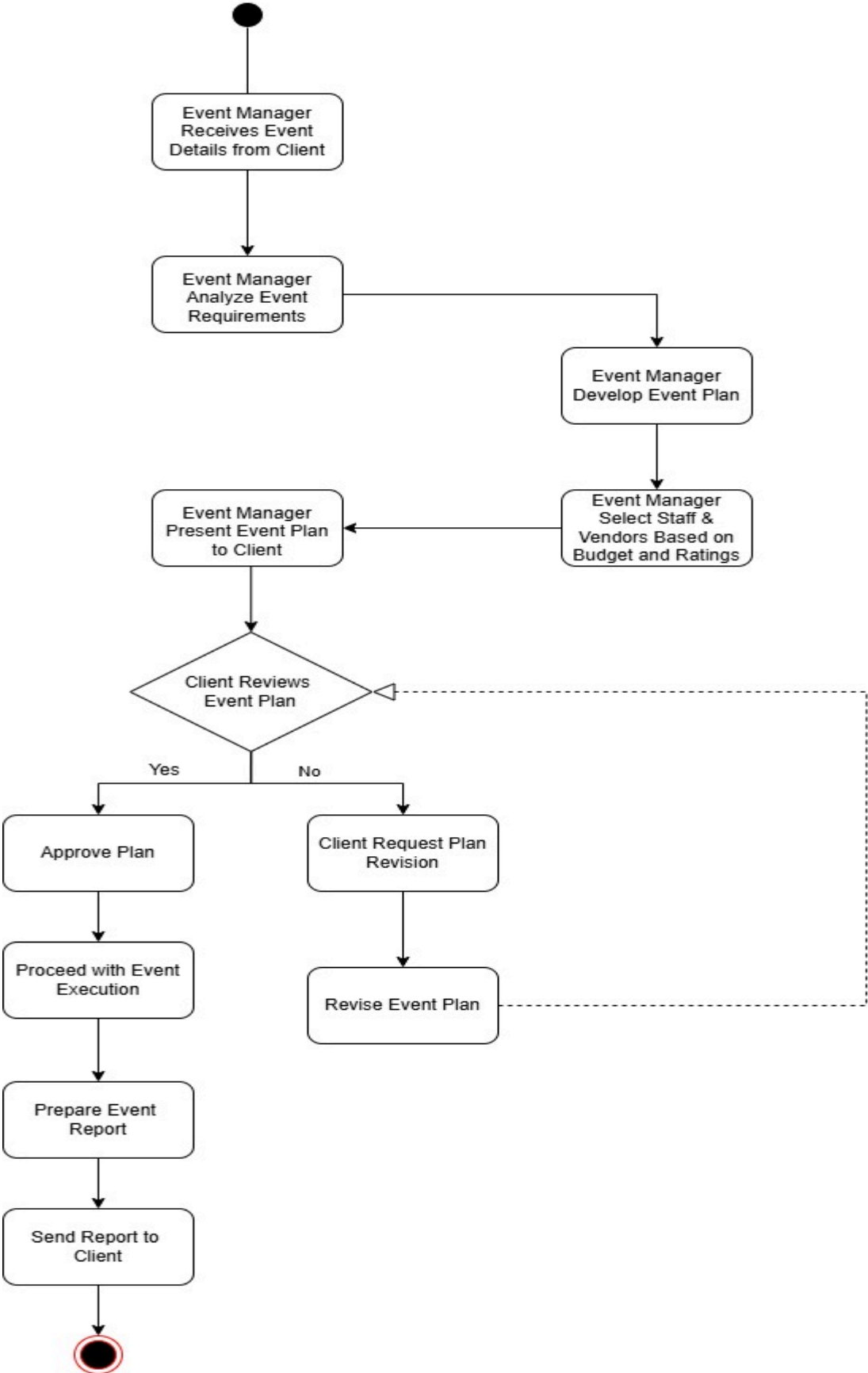
USE CASE Diagram



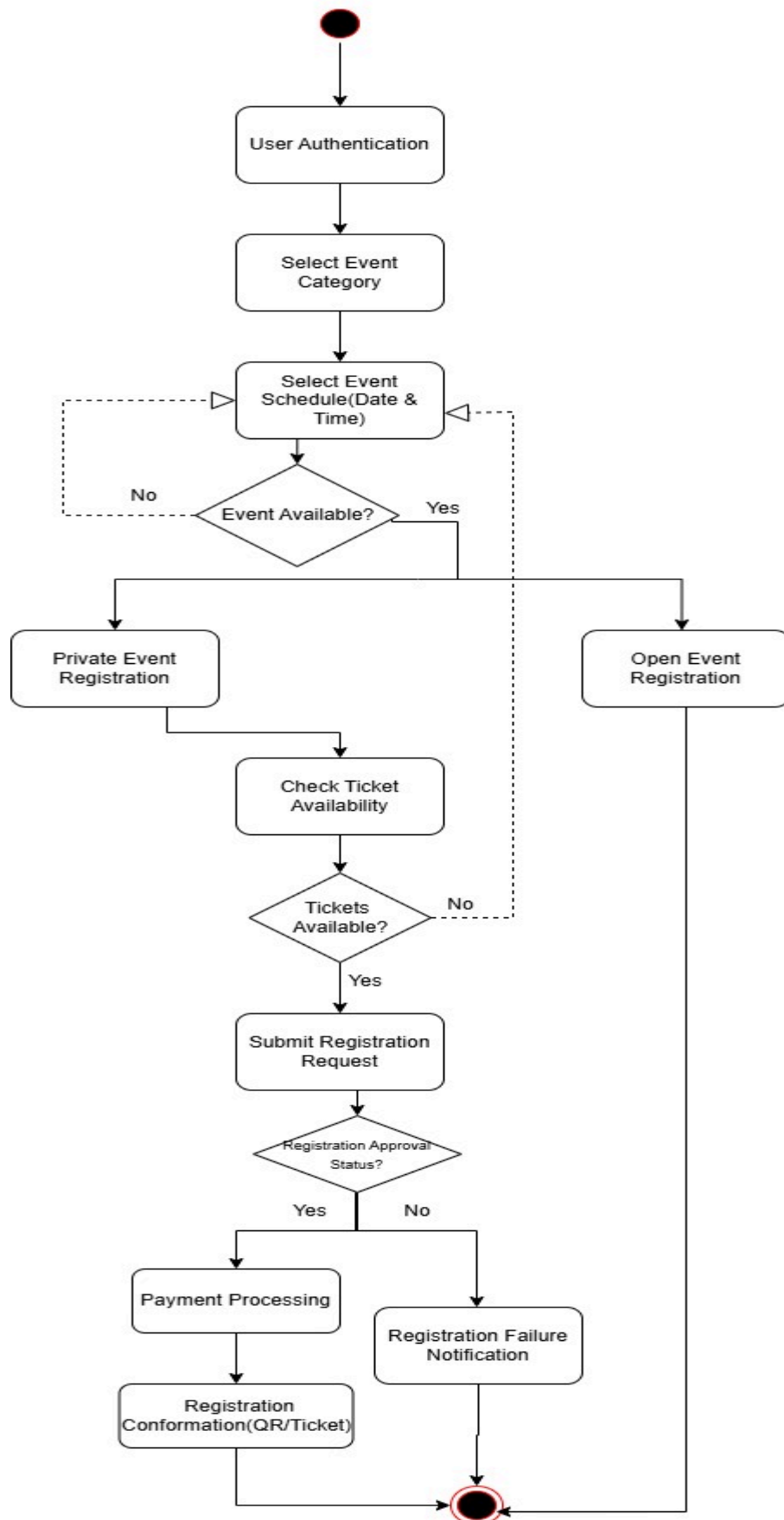
CLIENT REGISTRATION



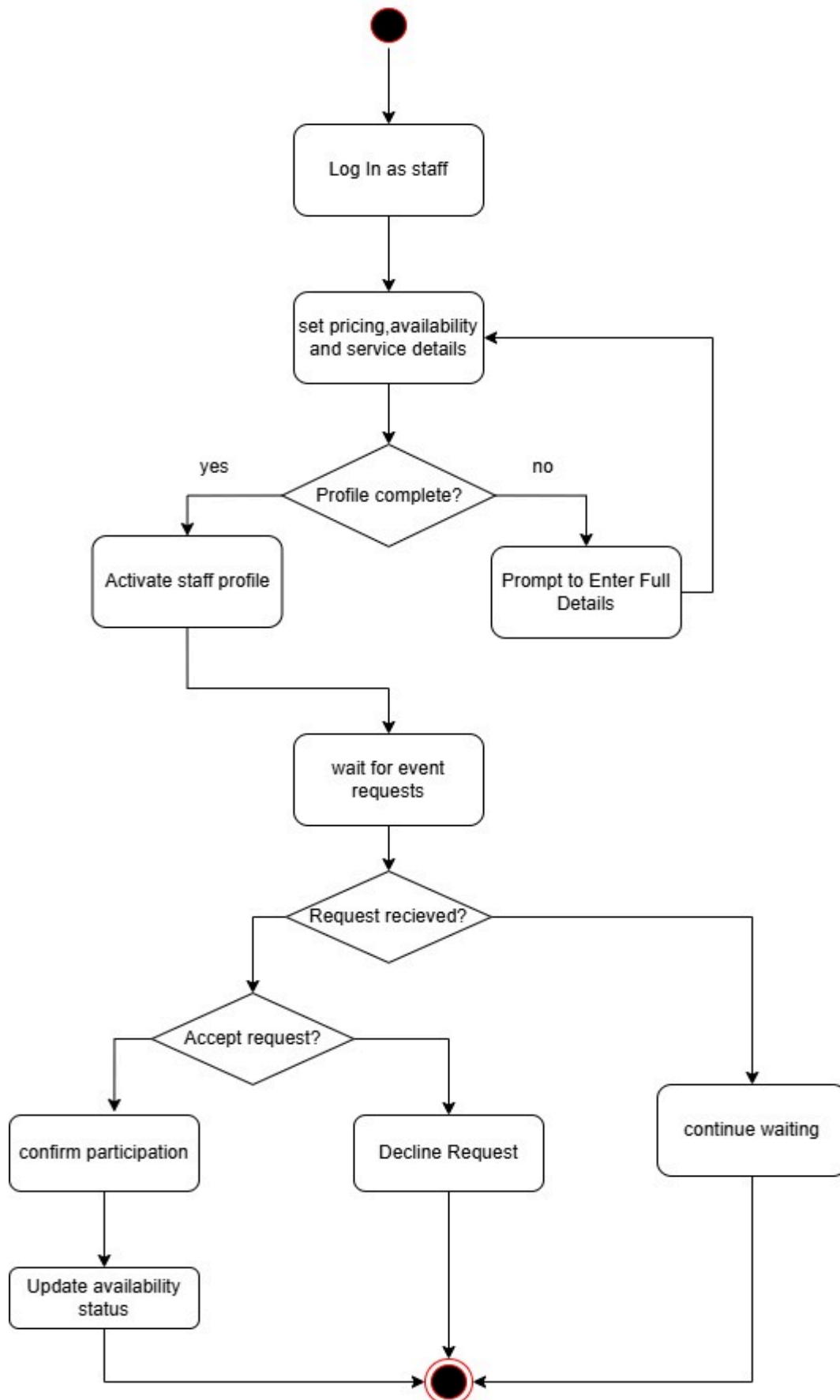
CLIENT - EVENT
MANAGER EVENT
APPROVAL



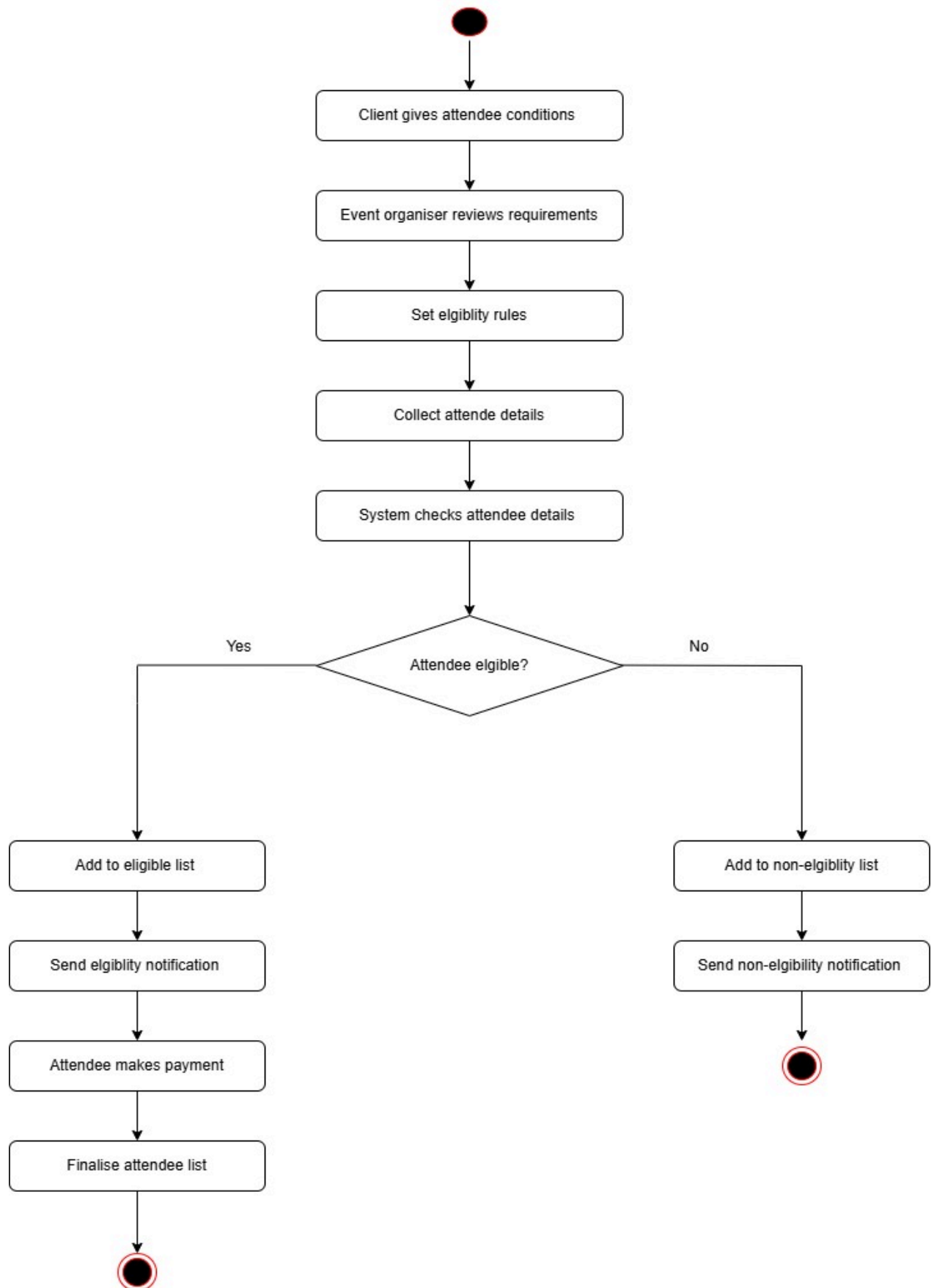
ATTENDEE REGISTRATION



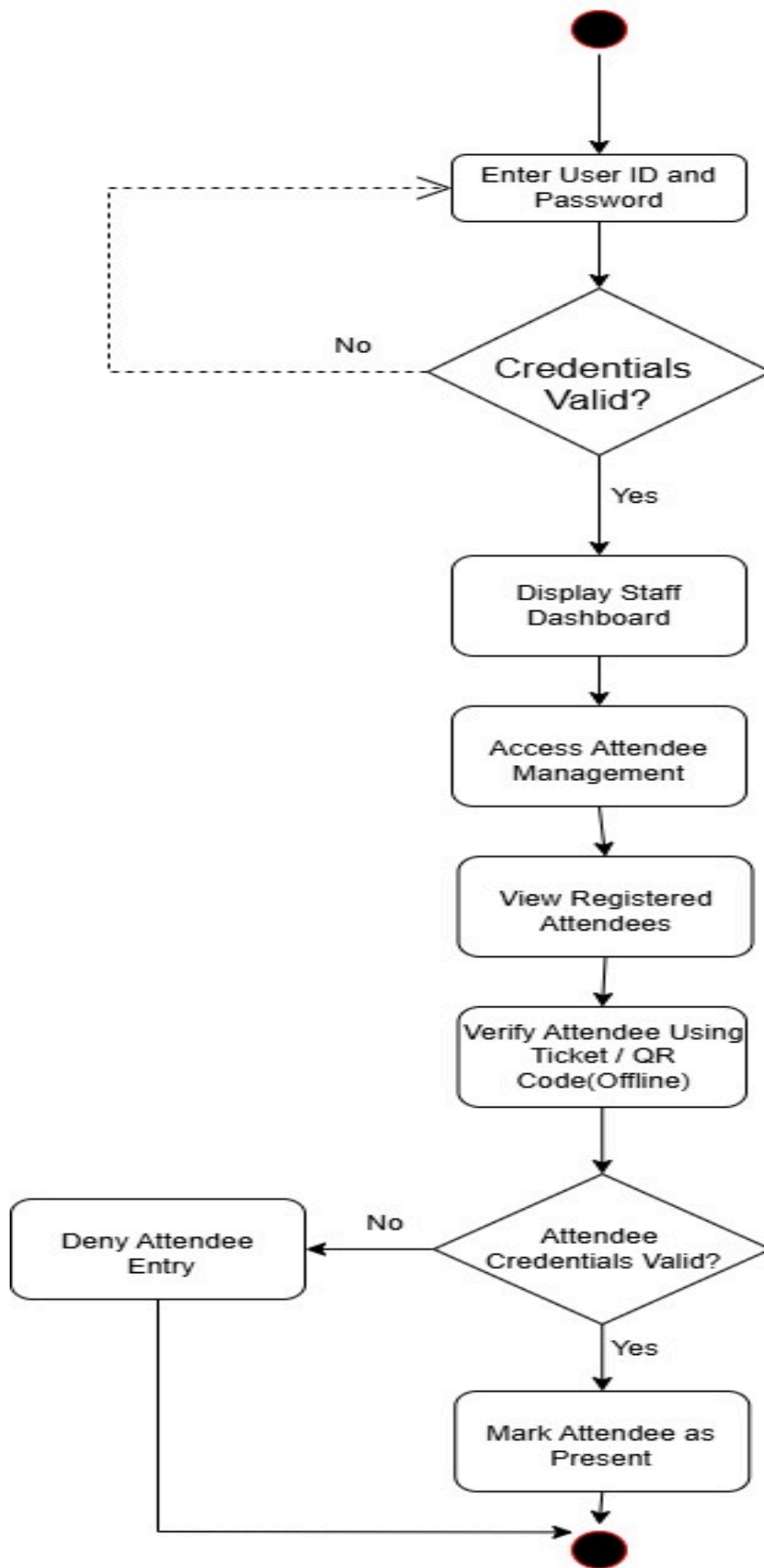
STAFF REGISTRATION

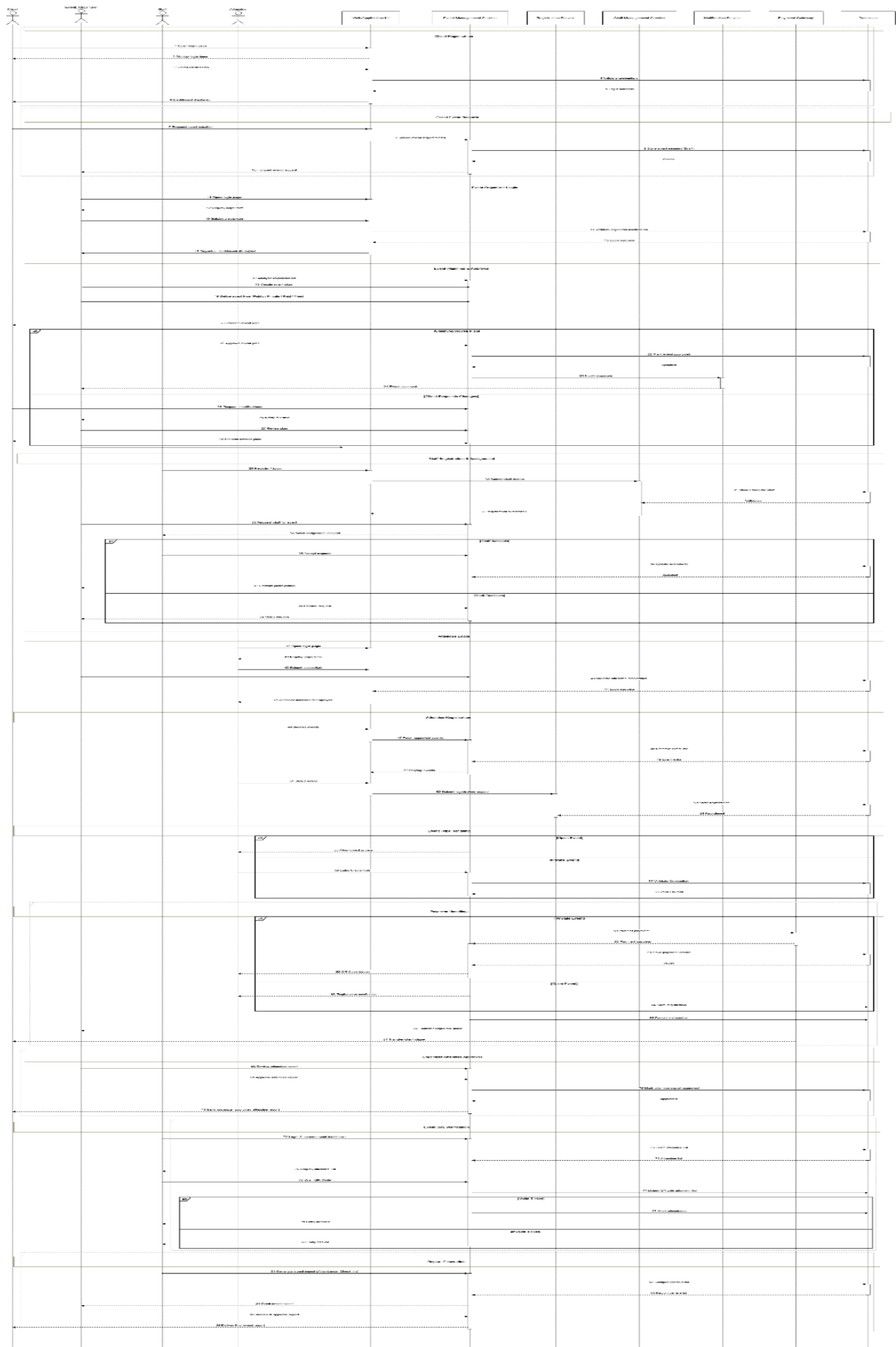


ATTENDEE MANAGEMENT



ATTENDEE VERIFICATION





UML Diagrams :

Use Case Diagram:

https://drive.google.com/file/d/1vRj5NivqbOisSu61z78ktw7Tp8ty1vV5/view?usp=drive_link

URL for Activity Diagrams:

<https://drive.google.com/drive/folders/1M3g5uHjMNVwoIXJLzsrHdOGoZGm78ZGE>

URL for sequence Diagram:

https://drive.google.com/file/d/14mOAkwDzvCYGvXOvTxI3m0qDMOVr1F07/view?usp=drive_link